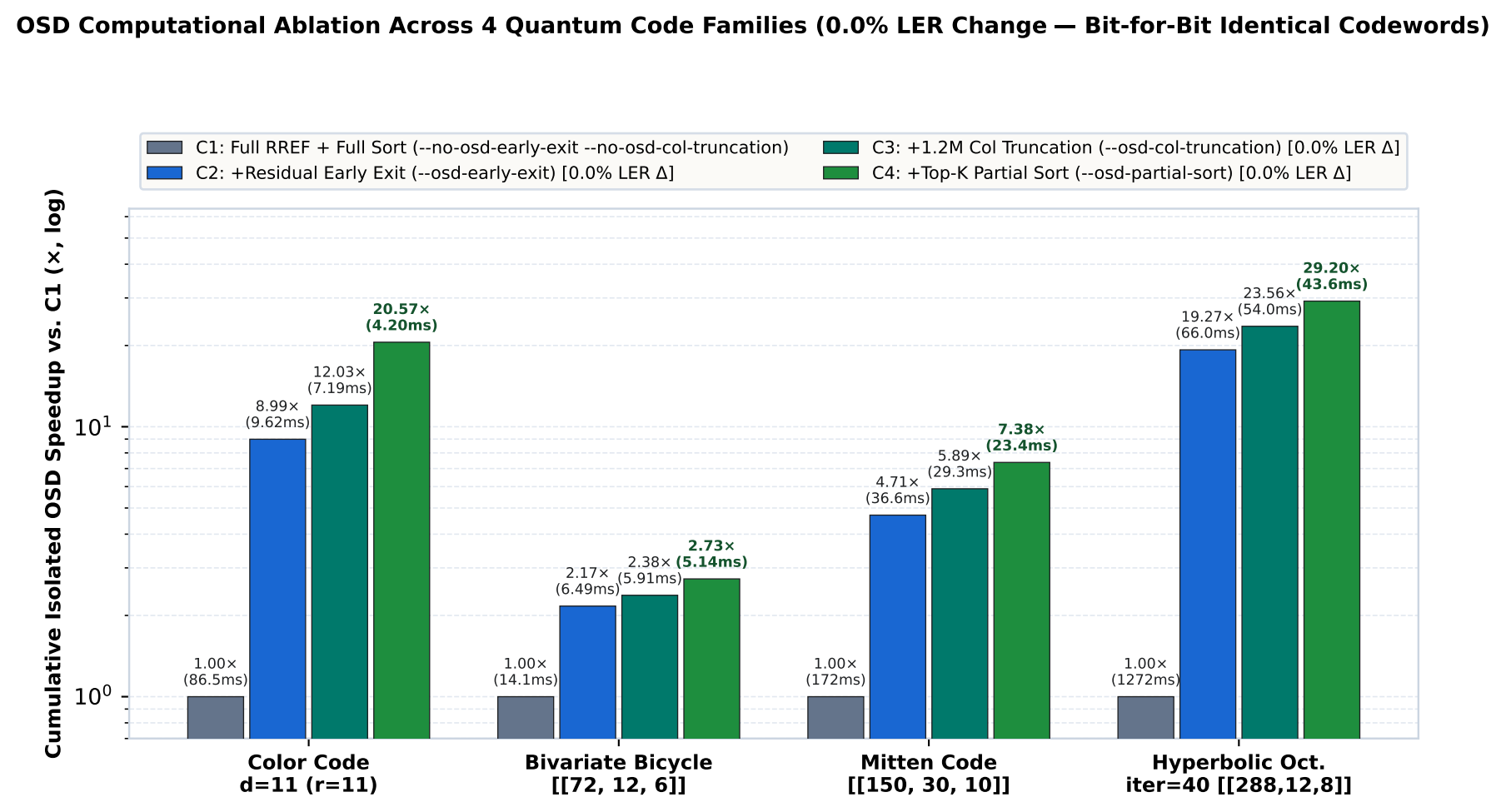
OSD Computational Ablation Across 4 Quantum Code Families (0.0% LER Change — Bit-for-Bit Identical Codewords)
C1: Full RREF + Full Sort (--no-osd-early-exit --no-osd-col-truncation)
C2: +Residual Early Exit (--osd-early-exit) [0.0% LER Δ]
C3: +1.2M Col Truncation (--osd-col-truncation) [0.0% LER Δ]
C4: +Top-K Partial Sort (--osd-partial-sort) [0.0% LER Δ]
Cumulative Isolated OSD Speedup vs. C1 (×, log)
10^1
10^0
1.00× (86.5ms)
8.99× (9.62ms)
12.03× (7.19ms)
20.57× (4.20ms)
1.00× (14.1ms)
2.17× (6.49ms)
2.38× (5.91ms)
2.73× (5.14ms)
1.00× (172ms)
4.71× (36.6ms)
5.89× (29.3ms)
7.38× (23.4ms)
1.00× (1272ms)
19.27× (66.0ms)
23.56× (54.0ms)
29.20× (43.6ms)
Color Code d=11 (r=11)
Bivariate Bicycle [[72, 12, 6]]
Mitten Code [[150, 30, 10]]
Hyperbolic Oct. iter=40 [[288,12,8]]